COURSE TITLE	**SUPPLY CHAIN ANALYTICS**
COURSE CODE	**04BA0602**
COURSE CREDITS	**4**

Objective:

1 N/A

Course Outcomes: After completion of this course, student will be able to:

1 Understand the components and significance of supply chain analytics in modern business environments.

2 Apply analytical tools to forecast demand and manage supply chain operations effectively.

3 Analyze inventory management problems and develop solutions using quantitative methods.

4 Explore the role of HR analytics in workforce planning and performance assessment within the supply chain context.

5 Identify and interpret trends in labor turnover and employee cost modeling using predictive analytics

Pre-requisite of course: Basic understanding of supply chain management and introductory statistics.

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
4	0	0	50	30	20	0	0

Contents : Unit	Topics	Contact Hours
1	**Foundations of Supply Chain Analytics** Components of Supply Chain Analytics, Role of Big Data in Supply Chain, , Strategic Goals and Demand Analytics, , Forecasting Techniques: Unbiased Forecast, Forecasting for Stable Products, Forecasting for New Replacement and Non-Replacement Products, , Measuring Forecast Accuracy	11
2	**Sales and Operations Planning Analytics** Introduction to Sales and Operations Planning, , Newsvendor Problem Analysis, Analyzing Demand Variability, , Supply and Resource Constraints, , Managing Excess Supply	9
3	**Inventory and Supply Optimization** Introduction to Inventory Management Problems, , Lot Size Restrictions and Volume Discounts, , Joint Replenishment Strategies, , Lead Time Variability, Dynamic Inventory Control Models	10

MR. PARAS CHANDULAL RUGHANI
Digitally signed by (Name of HOD)

DR. SANJEET SINGH
Digitally signed by (Name of Dean/ Principal)

Contents : Unit	Topics	Contact Hours
4	**HR Analytics in Supply Chain** Importance of HR Analytics in Supply Chain Management, Changing Dynamics of Workforce Management, , HR Analytics Framework, Workforce Planning and Sourcing Analytics, Acquisition/Hiring and Onboarding Analytics, Culture Fit and Engagement Analytics, , Performance Assessment and Employee Lifetime Value Modeling, , Employee Wellness and Health Analytics	16
5	**Trends and Predictive HR Analytics** Analyzing Turnover Trends and Labor Turnover In, Correlation Analysis in HR Metrics, Predictive Forecasting for Workforce Growth, Analyzing Employee Work Patterns and Prime Working Days, Employee Standard Unit Cost and Appraisal Cost Analysis	14
Total Hours		**60**

Textbook :

1 Supply Chain Analytics: Strategies, Models and Solutions. , Bhattacharya, R., & Bhattacharyya, A. M. , Sage Publications, 2025

2 Supply Chain Analytics. , Sanders, N. R. , Prospect Press, 2019

References:

1 Supply Chain Analytics and Modelling: Quantitative Tools and Applications., Supply Chain Analytics and Modelling: Quantitative Tools and Applications., Tipi, N. , Kogan Page., 2019

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation

Remember / Knowledge	Understand	Apply	Analyze	Evaluate	Higher order Thinking / Creative
0.00	0.00	35.00	35.00	30.00	0.00

Instructional Method:

1 THEORY

Supplementary Resources:

1 https://www.britannica.com/science/statistics/Random-variables-and-

MR. PARAS CHANDULAL RUGHANI
Digitally signed by (Name of HOD)

DR. SANJEET SINGH
Digitally signed by (Name of Dean/ Principal)